

Supplementary material

Polyconvex neural-network surrogates with learned paired directional features for anisotropic hyperelastic homogenization

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This supplement documents additional SC-RVE constitutive diagnostics, namely the out-of-domain response, sampled finite-strain curvature, and closed-cycle work. The cycle audit includes both learned constitutive laws and reduced microscopic models.

S1 Out-of-domain constitutive response

The available SC-RVE fitting data occupy the engineering-strain box

$$\mathbf{e}_{\min} = (0, -0.099, -0.168)^T, \quad \mathbf{e}_{\max} = (0.196, 0.016, 0.108)^T, \quad (\text{S1})$$

with $\mathbf{e} = (E_{11}, E_{22}, 2E_{12})^T$. The independent test set contains 400 states. An additional 350-state probe extends selected distances from the box center by factors 1.05, 1.10, 1.25, 1.50, and 2.00. Five probe states have no finite FOM label and are excluded with the same mask for all compared models. Validation stress selected every checkpoint before the test and probe responses were evaluated.

The Unconstrained energy, trained under the same protocol as the constrained models, has aggregate stress errors of 0.0052% on the test set and 0.668% on the finite probe set. It is more accurate than the selected six-feature models on both sets, so admissibility does not buy extrapolation accuracy on this cell. Thirty-two learned features, trained under the same protocol, extrapolate less accurately than six; Appendix F of the main paper reports them together with the capacity controls.

Table S1: SC-RVE: test and probe constitutive errors of the selected models, in percent.

Model	e_S^{test} [%]	e_W^{test} [%]	e_S^{probe} [%]	Worst test [%]	Worst probe [%]
ICNN	0.6214	0.1929	10.6433	4.9263	114.3506
ICKAN	0.6359	0.1900	10.0978	4.8240	106.1888
Unconstrained energy	0.0052	0.0010	0.6680	0.0203	6.0318

All values are percentages. The last two columns are maximum per-state stress errors; the first three are aggregate array norms.

Test: 400 states; probe: 345 finite reference states out of 350.

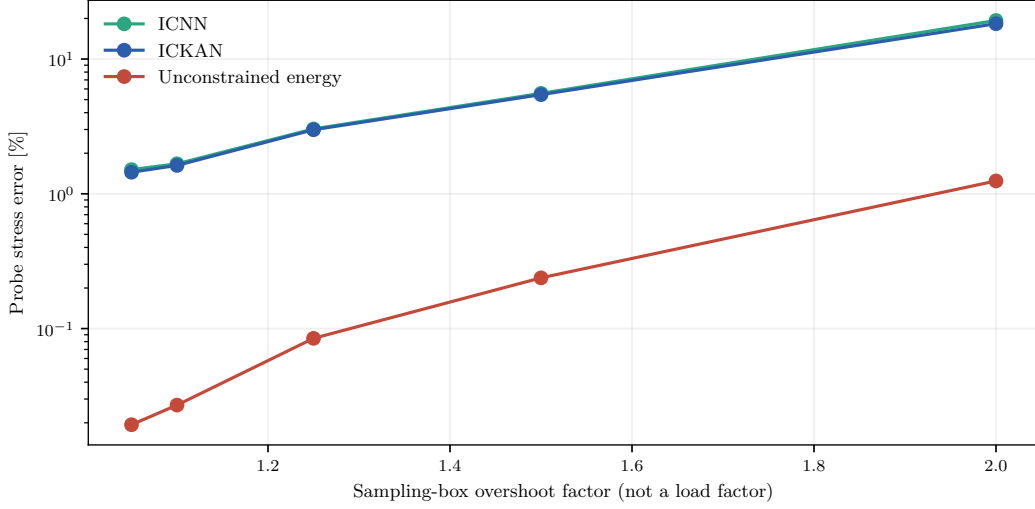


Figure S1: Stress error on increasingly distant probe rings, using only finite reference labels. Ring factors expand the sampling box; they are not coupon load multipliers. The constrained errors reach about 19% at factor 2, against 1.2% for the Unconstrained energy; admissibility does not imply extrapolation accuracy.

S2 Rank-one curvature inside and beyond the data

For $\mathbf{H} = \mathbf{a} \otimes \mathbf{b}$, the relevant finite-strain curvature is

$$D_{\mathbf{F}}^2 W[\mathbf{H}, \mathbf{H}] = (\delta \mathbf{e})^T \mathbf{D} \delta \mathbf{e} + \mathbf{S} : (\mathbf{H}^T \mathbf{H}), \quad \delta \mathbf{E} = \text{sym}(\mathbf{F}^T \mathbf{H}), \quad (\text{S2})$$

where $\delta \mathbf{e} = (\delta E_{11}, \delta E_{22}, 2\delta E_{12})^T$, $\mathbf{D}$ is the engineering-strain stress tangent, and $\mathbf{S}$ is the second Piola–Kirchhoff stress. Here $\mathbf{a}, \mathbf{b} \in \mathbb{R}^2$ define the rank-one direction and $\text{sym}(\mathbf{A}) = (\mathbf{A} + \mathbf{A}^T)/2$. Omitting the second term tests a different property. In particular, a positive semidefinite tangent with respect to $\mathbf{e}$ does not generally imply rank-one convexity in compression. For energy-based models, Eq. (S2) is evaluated over sampled states and unit directions; negative candidates are checked by second differences of $W(\mathbf{F} \pm h\mathbf{H})$.

Table S2: Lowest sampled rank-one curvature inside and beyond the data.

State set	N	Unconstrained	ICNN	ICKAN
Held-out test	400	200.8	224.3	211.2
In-box states	12 000	181.3	218.3	198.9
Finite-label probe	345	135.0	200.0	186.8
Radial expansion, up to 4× the box	109 743	−362.5	183.1	176.7
Principal stretches 0.25–3	24 000	−384.9	186.3	185.4

Minimum sampled rank-one curvature in MPa, over 180 unit directions $\mathbf{b}$ with exact minimization over $\mathbf{a}$. Minima need not occur at the same state or direction. The last two sets lie beyond the data.

Table S2 lists the lowest sampled curvature of each energy on five state sets. On the held-out test, additional in-box, and finite-label probe states, all three energies have positive sampled curvature. Beyond the data, a radial expansion of the fitting box and a broad scan of principal stretches find negative curvature for the Unconstrained energy at 14 108 of 109 743 and 10 048 of 24 000 states, respectively. Along the radial expansion, the first negative value appears at 1.55 times the box, with $J = 0.780$ and curvature −12.4 MPa. Second differences of $W(\mathbf{F} \pm h\mathbf{H})$ at $h = 10^{-4}$, 3×10^{-4} , and 10^{-3} reproduce the most negative values to within 10^{-5} relative. The ICNN and ICKAN have positive curvature on every set, as their analytical polyconvexity argument requires; their samples check the implementation.

S3 Closed-cycle work of the effective stress

For a hyperelastic law, returning to the same deformation on the same reversible equilibrium branch returns the stored energy to its initial value. The net stress work must therefore vanish:

$$\mathcal{W}_{\text{cyc}} = \oint \mathbf{s} \cdot d\mathbf{e} = 0, \quad \mathbf{e} = (E_{11}, E_{22}, 2E_{12})^T, \quad \mathbf{s} = (S_{11}, S_{22}, S_{12})^T. \quad (\text{S3})$$

Here $\mathbf{e}$ and $\mathbf{s}$ are work-conjugate strain and stress vectors. A nonzero result beyond numerical error rules out an exact potential for that stress map; a zero result on one cycle cannot establish one. The test does not assess polyconvexity.

We use a rectangle of half-width 0.001 in (E_{11}, E_{22}) , with fixed shear and center

$$\mathbf{e}_c = (0.0051260033, -0.0945507613, -0.1283720602)^T.$$

It lies inside the SC-RVE fitting box. The center was fixed in an earlier audit and is reused unchanged for every model.

Table S3: Net work around the counterclockwise SC-RVE cycle, in J m^{-3} .

Model	$\mathcal{W}_{\text{cyc}}$
FOM (8 per edge)	-6.98×10^{-10}
Unconstrained energy	-9.20×10^{-11}
ICNN	4.02×10^{-10}
ICKAN	1.09×10^{-5}
HPROM	-1.23×10^0
HPROM-ANN	2.48×10^2
D-HPROM-ANN	5.41×10^1

128 Gauss points per edge, except for the FOM. The ICKAN residual decreases with quadrature refinement through its spline knots.

The Unconstrained energy, ICNN, and ICKAN obtain stress by differentiating an energy; their residuals are consistent with numerical integration and roundoff. The deployed HPROM outputs retain nonzero work under quadrature refinement. Reversing the cycle reverses its sign, and tighter equilibrium tolerances leave the intrusive residuals essentially unchanged.

The HPROMs fit equilibrium and stress cubatures separately, so their effective stresses are not guaranteed to derive from one energy. This diagnostic concerns those implementations. Among them, the largest ratio of absolute net work to accumulated absolute work along this cycle is 0.0205%. The selected cycle establishes neither a general ranking of the models nor the frequency of such defects across the strain domain.

S4 Training and implementation details

The initial paired features are selected from a bank of 2028 admissible candidates. The bank combines twelve orientations $\theta_i = k\pi/12$ with the stretch powers $p_i, q_i \in \{0.5, 1, 1.5, 2, 3\}$ and with determinant exponents at the bounds and midpoint of $0 \leq b_i \leq 2p_i - 1$ and $0 \leq c_i \leq 2q_i - 1$, moved slightly inside these intervals so that the strict exponent parameterization can represent them. Nonnegative least-squares fits of the fitting response and of the reference tangent identify the contributing candidates. The largest contributions are retained when more than m are selected, and pivoted QR completes the set when fewer are.

Adam starts from learning rates of 5×10^{-3} for the ICNN and ICKAN and 4×10^{-4} for the Unconstrained energy, halved after 40 validation checks without improvement down to 10^{-5} , with a check every ten steps. It stops when the best validation error improves by less than 0.1% over 100 checks at the smallest learning rate. L-BFGS, in calls of at most ten iterations, stops under the same criterion over 30 calls. Each model is trained from the seeds 16, 29, and 47, and the SC-RVE fitting and validation states are split with seed 5. On the SC-RVE, validation selects seed 16 for both constrained energies and seed 47 for the Unconstrained energy.